

Academic Profile of Prof. Georgios Neofotistos

Since August 2026, Professor Georgios Neofotistos has been with the Faculty of Engineering at the Tashkent University of Technology. He is also [Associate in Applied Physics](#) at the John A. Paulson School of Engineering and Applied Sciences (SEAS) at Harvard University, where he has served as Visiting Scholar and Visiting Lecturer since 2017, and Affiliated Researcher at the Institute of Theoretical and Computational Physics at the University of Crete (Greece), where he has served as Adjunct Professor in the Physics Department. His career uniquely bridges theoretical and applied physics, computational science, machine learning, and science education across institutions in the United States, Greece, and beyond, making him one of the most versatile academic figures in his field.

At Harvard, he has established himself as both an outstanding educator and an active researcher. He has co-taught, with Harvard faculty, courses including the APMTH (Applied Math) 101 *Statistical Inference for Scientists and Engineers*, APMTH 10 *Computing with Python for Scientists and Engineers*, PHYS 20 *Introduction to Computational Physics*, and has been a key member of the teaching staff for APMTH 104 *Complex Analysis and Series Expansions for Applications to Science, Engineering and Finance*. His dedication to teaching has been formally recognized by Harvard: he was awarded Harvard's Certificate of Distinction in Teaching granted by the Derek Bok Center for Teaching and Learning in cooperation with the Office of Undergraduate Education for his work in both the APMTH 104 (Fall 2018) and the APMTH 10 course in (Fall 2019). Most recently, in the Fall semester of 2024, he achieved a perfect overall evaluation score of 5.0 out of 5.0 in Harvard Faculty of Arts and Sciences course evaluations, a testament to the exceptional quality and impact of his teaching.

His research collaboration at Harvard focuses on the frontier of machine learning applied to complex physical systems and biomedical data. His contributions include the development of novel machine learning frameworks using neural networks and sparse sensing to predict complex spatiotemporal dynamics, as well as extensive work on epidemic modeling during the COVID-19 pandemic. Together with Harvard colleagues, he co-authored some of the widely read early studies on COVID-19 epidemic wave modeling, including studies on multiple wave epidemics, the effect of vaccination campaigns, and the susceptibility to resurgent outbreaks. This work exemplifies his ability to translate advanced computational methods into real-world, high-stakes applications.

At the University of Crete, where he has been affiliated since 1999, Professor Neofotistos taught courses in machine learning, deep learning and data science, and has for many years taught courses ranging from mathematical finance and global climate change to high-tech entrepreneurship and complex networks science. He played a foundational role in building the University of Crete's innovation and entrepreneurship ecosystem, serving as coordinator of its Innovation and Entrepreneurship Center and its Entrepreneurship Programme. He also represented the University on the Regional Innovation Council of Crete and organized numerous European COMENIUS-GRUNDTVIG summer schools for innovative science educators from across Europe on topics including inquiry-based science education, computational thinking, and open science resources. His current research at the University of Crete focuses on developing classical-quantum pipelines in latent spaces.

Prof. Neofotistos received his Ph.D. in Physics from Temple University (Philadelphia, USA), with a dissertation on quantum transport in submicron semiconductor devices. He got his M.A. in Physics from Temple University, and his B.Sc. in Physics from the University of Athens (Greece).

After getting his Ph.D., he conducted post-doctoral research at Purdue University's Department of Electrical Engineering. He served as visiting assistant professor at Temple University's Physics Department (teaching the graduate solid-state course and the undergraduate statistical mechanics course), and as assistant professor at the Physics and Space Sciences Department of the Florida Institute of Technology (Florida Tech) in Melbourne, Florida, USA, teaching undergraduate science courses.

Prof. Neofotistos's scientific publications span quantum transport physics, complex network theory, machine learning for dynamical systems, COVID-19 modeling, educational innovations, and science education policy. His work has appeared in leading journals including *Physical Review B*, *Frontiers in Physics (Quantum Engineering and Technology)*, *Physics Letters A*, *Chaos Solitons and Fractals*, *IEEE Transactions on Electron Devices*, *Journal of Medical Internet Research*, *Viruses*, and *AI in Medicine* among others. He has been a contributor to major international collaborative projects and brings to his work a rare combination of deep theoretical grounding, computational expertise, and a long-standing commitment to educational innovation at the highest institutional levels.

Selected publications in peer-reviewed journals:

Addressing Class Imbalance in ECG Arrhythmia Classification Using Latent Diffusion and Quantum-Enhanced Generative Modeling, by G. Kritopoulos, G. Neofotistos, G. Barmparis, and G. Tsironis, *AI in Medicine*, 1(3); <https://doi.org/10.3390/aimed1030023> (2026).

Susceptibility to Resurgent COVID-19 Outbreaks Following Vaccine Rollouts: A Modeling Study, by Neofotistos, G.; Angeli, M.; Mattheakis, M.; Kaxiras, E. *Viruses* 2022, 14, 2237. <https://doi.org/10.3390/v14102237>

Modeling the effect of the vaccination campaign on the Covid-19 campaign, by Angeli M., Neofotistos G., Mattheakis M., Kaxiras E., *Chaos, Solitons & Fractals* 154 (2022) 111621.

Robust prediction of complex spatiotemporal states through machine learning with sparse sensing, by Barmparis G, Neofotistos G, M.Mattheakis, Hitzanidi J, Tsironis GP, Kaxiras E., *Physics Letters A*. 384 (24) 126300 (2020), <https://doi.org/10.1016/j.physleta.2020.126300>.

Multiple Epidemic Wave Model of the COVID-19 Pandemic: Modeling Study, by Kaxiras E, Neofotistos G, *J Med Internet Res* 2020;22(7):e20912, DOI: 10.2196/20912, PMID: 32692690, PMCID: 7394522.

The first 100 days: Modeling the evolution of the COVID-19 pandemic, by Kaxiras E, Neofotistos G, Angelaki E. *Chaos, Solitons & Fractals* 138 (2020) 110114.

Machine Learning With Observers Predicts Complex Spatiotemporal Behavior, by Neofotistos G, Mattheakis M, Barmparis GD, Hizanidis J, Tsironis GP and Kaxiras E, *Frontiers in Physics*, 7:24 (2019), doi: 10.3389/fphy.2019.00024.

Chimera states and synchronization in magnetically driven SQUID metamaterials, by J. Hizanidis, N. Lazarides, G. Neofotistos, and G. P. Tsironis *European Physical Journal Special Topics* 225, 1231-1243 (2016).

Chimeras in SQUID metamaterials, by N. Lazarides, G. Neofotistos, and G. P. Tsironis, *Physical Review B* 91, 054303 (2015).

Engineering and Design as Most Central Knowledge in the Architecture & Engineering Jobs, by P. G. Asteris, and G. C. Neofotistos, *Journal of Leadership and Management in Engineering; American Society of Civil Engineers (ASCE)*, LMENG-146, issue of January 2012.

Quantitative Analysis of the Usage of the COSMOS Science Education Portal, by S. Sotiriou, F. X. Bogner, and G. Neofotistos, *Journal of Science Education and Technology*. DOI: 10.1007/s10956-010-

9256-1. Springer Science+Business Media, LLC 2010. Accessible online at:

<http://www.springerlink.com/content/776g586483203156/>

Inelastic scattering effects on single barrier tunneling, by G. Neofotistos, R. Lake, and S. Datta, *Physical Review B* 43, 2442 (1991).

Quantum mechanical simulation of charge transport in very small semiconductor structures, by M. C. Yalabik, G. Neofotistos, K. Diff, H. Guo, and J. D. Gunton, *IEEE Transactions on Electron Devices* 36, 10009 (1989).

Resonant tunneling in double-barrier parabolic well structures, by G. Neofotistos, H. Guo, K. Diff, and J. D. Gunton, *IEEE Transactions on Electron Devices* 36, 745 (1989).

Time-dependent investigation of the resonant tunneling in a double-barrier quantum well, by H. Guo, K. Diff, G. Neofotistos, and J. D. Gunton, *Applied Physics Letters* 53, 131 (1988).

Simulation of charge transport in a GaAs MESFET using the time-dependent Schroedinger equation, by M. C. Yalabik, J. D. Gunton, G. Neofotistos, and K. Diff, *Superlattices and Microstructures* 3, 463 (1987).